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Mapping and predicting urban heat island intensity hotspots through a space–time machine learning framework in Bangladesh

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Abstract

Urban Heat Island Intensity (UHII) has become a pressing urban climate issue in rapidly developing nations such as Bangladesh. This study presents a nationwide thana-level assessment of UHII trends from 1990 to 2023 using remote sensing, geospatial analysis, and machine learning techniques. Land Surface Temperature (LST) was derived from Landsat imagery to quantify UHII, and a Space–Time Cube framework with Mann–Kendall trend analysis was applied to identify persistent, intensifying, emerging, and diminishing hotspot patterns. Major urban centers, including Dhaka, Narayanganj, and Khulna, exhibited increasing UHII, while Barisal and Mymensingh showed emerging cold spots. A Random Forest model was developed to forecast UHII up to 2040, revealing further intensification in densely populated and industrial zones. The results indicate that major metropolitan areas, particularly Dhaka, Narayanganj, and Khulna, exhibit persistent and intensifying heat hotspots, whereas divisions like Barisal and Mymensingh show emerging cold spots. The findings emphasize the need for climate-responsive urban planning and green infrastructure. This study establishes a baseline for long-term UHII monitoring and serves as a framework for future research aimed at developing predictive models and targeted mitigation strategies to enhance urban climate resilience.

Keywords Urban heat island intensity, Land surface temperature, Remote sensing, Machine learning, Space time cube, Emerging hotspot analysis

1 Introduction

Urbanization is the defining trend of the twenty-first century, with over half of the global population residing in urban areas, a proportion expected to reach 68% by 2050 (Chen et al., 2017; United Nations, 2019). Rapid urban expansion, particularly in developing regions such as South Asia, is primarily driven by population growth and land use changes (Kotharkar et al., 2018). While urbanization fosters economic development and infrastructure

growth, it also significantly alters natural landscapes, leading to environmental consequences, including local and regional climate modifications (Girardet, 2020). Among these, the Urban Heat Island (UHI) effect, which is the temperature variation between urban and rural areas, has been one of the most extensively studied in the urban climatology discipline (Heisler & Brazel, 2010). First documented in London in the early nineteenth century (Howard, 1833), UHI has since been observed worldwide, driven by the replacement of natural surfaces with heat-absorbing materials such as concrete and asphalt (Karakuş, 2019; Kaya et al., 2012). The Urban Heat Island Intensity (UHII) quantifies this phenomenon, representing the temperature difference between urban and surrounding rural environments (Peng et al., 2012; Santamouris, 2015). There is a clear distinction between

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the Urban Heat Island (UHI) and Urban Heat Island Intensity (UHII). UHI refers to the general phenomenon of urban–rural thermal differences, while UHII denotes the specific, quantifiable temperature difference (ΔT) between urban areas and their rural surroundings. UHII is influenced by various factors, including population density, height-width ratio of the built-up area characteristics, climatic conditions, and anthropogenic heat emissions (Manoli et al., 2019; Tzavali et al., 2015; Wang et al., 2016; Zhang et al., 2022).

Urbanization-induced temperature rise exacerbates heat stress, air pollution, and energy demand, disproportionately impacting densely populated cities (Luo & Lau, 2018; Santamouris, 2020). Extreme heat exposure already affects 30% of the world's population and is projected to impact nearly 48% by 2100, significantly increasing the risk of heat-related mortality (Mora et al., 2017). Vulnerable populations, including the elderly, outdoor workers, and individuals with pre-existing health conditions, face higher risks from prolonged exposure to UHI effects (Chakraborty et al., 2019; Heaviside et al., 2017). UHI not only affects human health but also intensifies energy consumption by increasing cooling demands, contributing to carbon emissions and urban overheating (Santamouris, 2015; Uddin et al., 2022). Furthermore, UHI-induced warming influences socio-economic conditions, reducing outdoor thermal comfort, labor productivity, and capital efficiency while straining energy infrastructure (Leal Filho et al., 2021; Shahfahad et al., 2022). Given these concerns, understanding UHI dynamics is essential for sustainable urban planning and climate adaptation strategies.

To characterize UHII, researchers have traditionally relied on two primary methodological approaches: in-situ field measurements and remote sensing-based assessments. In-situ methods, such as meteorological stations and mobile traverses, provide high-precision temperature data (Dewan et al., 2021; Hu et al., 2016). However, these methods face limitations in spatial coverage, instrument inconsistencies, and data gaps due to infrastructure constraints (Wang et al., 2007). To overcome these challenges, remote sensing has emerged as a robust alternative, enabling large-scale monitoring of land surface temperature (LST) variations using thermal sensors aboard satellites such as Landsat, MODIS, and Sentinel (Huang et al., 2021; Kerr et al., 2004; Lai et al., 2018). Numerous studies have demonstrated the relationship between land use and land cover (LULC) changes and LST trends, revealing how urban expansion contributes to thermal disparities (Abdullah et al., 2022; Fatemi & Narangifard, 2019; Gohain et al., 2021; Imran et al., 2021). Advanced cloud-computing platforms, such as Google Earth Engine (GEE), have further revolutionized

large-scale UHI analysis, offering computational efficiency, automated data retrieval, and multi-temporal analytics for UHI monitoring (Gadekar et al., 2023; Zhao et al., 2021).

In many developing countries, the intensifying Urban Heat Island Intensity (UHII) and deterioration of urban environmental quality can severely disrupt healthy lifestyles for millions, especially those living in densely populated and poorly planned urban areas (Tong et al., 2022). Bangladeshis particularly vulnerable to climate change-induced temperature variations (Eckstein et al., 2019). The country's landscape, characterized by low-lying floodplains and tropical monsoon conditions, experiences pronounced seasonal fluctuations in temperature and precipitation (Brammer, 2012). Urban temperature rise in Bangladesh has been linked to elevated cooling energy demands, deteriorating air quality, and increased heat stress, particularly affecting low-income communities (Akter et al., 2021; Rahman et al., 2021). Climate projections indicate that heatwaves will become more frequent in Bangladesh's metropolitan areas, exacerbating urban health and economy (Mourshed, 2011). While the country has implemented policies to mitigate flooding and natural disasters, urban heat mitigation remains under-addressed in climate adaptation strategies (GED, 2018).

Existing studies on UHI in Bangladesh have primarily focused on major urban centers such as Dhaka, Chattogram, and Rajshahi (Uddin et al., 2022; Kafy et al., 2019; Kafy et al., 2021; Jahan, 2024). Furthermore, most studies rely on static LST analyses, often using single-date satellite imagery or short-term observations that fail to capture long-term UHII trends (Akter et al., 2021; Uddin et al., 2022). Another critical research gap is the lack of predictive modeling for UHI trends, particularly using machine learning approaches. While machine learning has been widely employed for land-use modeling and environmental studies globally (Avdan & Jovanovska, 2016; Pande et al., 2024), its application in UHI forecasting remains largely unexplored in Bangladesh.

To address these gaps, this study presents the first thana-wise, nationwide assessment of Urban Heat Island Intensity (UHII) trends in Bangladesh from 1990 to 2023. It is also the first to predict future UHII trends up to 2040 using machine learning, specifically the Random Forest (RF) model. The novelty of this research does not lie in methodological innovation, as techniques such as LST retrieval, NDVI/NDBI-based classification, Emerging Hotspot Analysis (EHSA), and RF forecasting are widely established. Instead, its contribution lies in the systematic integration of these approaches and their application at a fine-scaled thana level across the entire country, which has not been

previously attempted. This framework emphasizes scalability, forecasting utility, and direct policy relevance, making it adaptable to other regions facing similar urban heat challenges. Additionally, this research introduces a classification framework for UHII trends, categorizing heat patterns into persistent, intensifying, emerging, and declining clusters, providing granular insights into spatial UHI variations. By leveraging GEE-based LST retrieval and RF classification, this study advances urban climatology research in Bangladesh and supports evidence-based climate adaptation policies.

2 Study area

Bangladesh, located in South Asia, extends between latitudes $20^{\circ}34'N$ to $26^{\circ}38'N$ and longitudes $88^{\circ}01'E$ to $92^{\circ}41'E$. It is bordered by India to the west, north, and east, Myanmar to the southeast, and the Bay of Bengal to the south (Fig. 1). Spanning roughly 147,570 square kilometers, the country is situated within a tropical monsoon climate zone, marked by elevated temperatures, high humidity, and clearly defined wet and dry periods (Brammer, 2012). By 2021, its population had reached approximately 166.5 million, accounting for about 2.11% of the global total (Khatun et al., 2021). Over the past few decades, rapid urbanization has reshaped

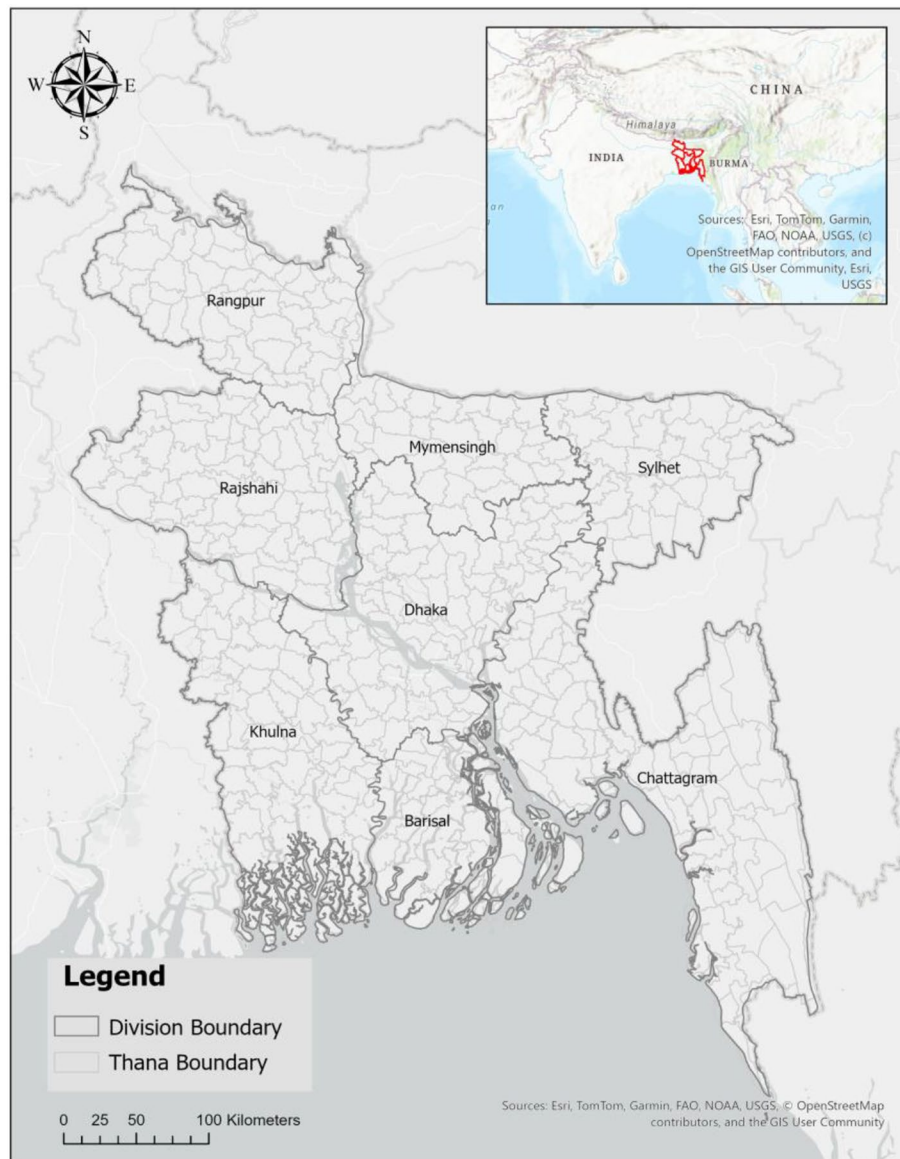


Fig. 1 Detailed map of the study area (Bangladesh) with its administrative boundaries

major metropolitan centers, such as Dhaka, Chattogram, Khulna, and Rajshahi, leading to extensive land use and land cover (LULC) transformations that have intensified the Urban Heat Island (UHI) effect (Rahman et al., 2021).

Administratively, Bangladesh is structured into divisions, districts, upazilas (sub-districts), and thanas (urban administrative units). This study focuses on Thanas, which represent the smallest administrative divisions within urban areas. Thanas serve as micro-level governance units, making them particularly relevant for assessing intra-urban variations in Land Surface Temperature (LST) and UHI intensity. Urban microclimates vary significantly within a single city due to factors such as vegetation cover, impervious surfaces, and population density, all of which influence localized temperature variations (Haque et al., 2019). Given their role in policy implementation and resource allocation, Thanas provide an ideal unit for analyzing urban climate dynamics and designing climate adaptation strategies. By employing Thanas as the spatial unit of analysis, this study ensures a higher degree of spatial granularity, which is often lost in broader-scale district- or city-level assessments of UHI effects.

3 Methods

This methodological flowchart (Fig. 2) outlines the subsequent steps undertaken in the study, from data acquisition and preprocessing to spatial-temporal analysis and predictive modeling. The workflow integrates geospatial techniques, machine learning, and statistical analysis to assess Urban Heat Island Intensity (UHII) dynamics and predict future trends up to 2040.

3.1 Data acquisition and preprocessing

Landsat 5 Thematic Mapper (TM) data were utilized for the period up to 2012, while Landsat 8 Operational Land Imager (OLI) data were employed for the years 2013 to 2023. Given the long temporal span of the study (1990–2023), Landsat data were selected as the most suitable option due to their consistent data availability and long-term archive. Each image was filtered to exclude scenes with cloud cover exceeding 10%, thereby minimizing atmospheric interference and ensuring

data reliability. Standard atmospheric correction techniques were applied to enhance radiometric accuracy and comparability across temporal datasets. The Landsat imagery, with a spatial resolution of 30 m, provided a balance between detail and coverage, making it well-suited for analyzing urban heat island effects over an extended period. The study area was delineated based on administrative Thana boundaries, ensuring a consistent spatial framework and uniform geographic representation throughout the study period.

3.2 Calculation of normalized difference vegetation index (NDVI)

NDVI was computed to assess vegetation cover using the red and near-infrared (NIR) bands:

$$\text{NDVI} = \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}} \quad (1)$$

where:

NIR: Reflectance in the near-infrared band.

Red: Reflectance in the red band.

This index ranges from -1 to 1 , with higher values indicating denser vegetation. The NDVI calculation methodology is well-documented in the literature (Pettorelli et al., 2005).

3.3 Calculation of normalized difference built-up index (NDBI)

NDBI was computed to assess built-up and impervious surfaces using the shortwave-infrared (SWIR) and near-infrared (NIR) bands:

$$\text{NDBI} = \frac{\text{SWIR} - \text{Red}}{\text{SWIR} + \text{Red}} \quad (2)$$

where:

SWIR: Reflectance in the shortwave-infrared band.

NIR: Reflectance in the near-infrared band.

This index ranges from -1 to 1 , with higher values indicating built-up or impervious surfaces, while lower values correspond to vegetated or water-covered areas. The

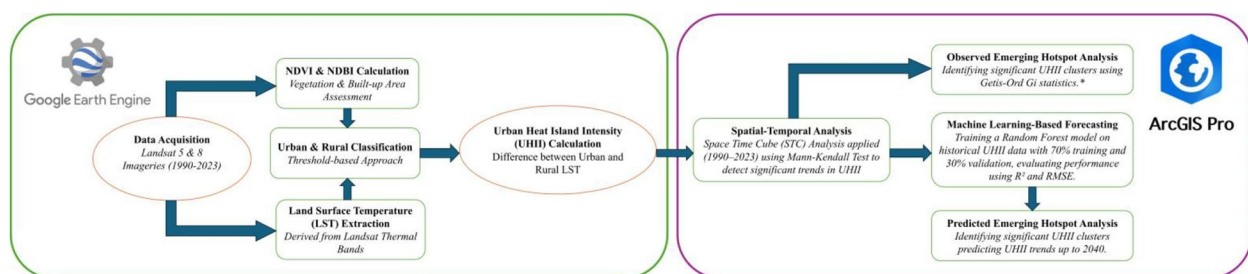


Fig. 2 Methodological flowchart illustrating the sequential steps of the analysis process

NDBI calculation methodology is well-documented in the literature (Zha et al., 2003).

3.4 Urban and rural area classification

Urban areas were delineated using threshold values of the normalized difference vegetation index (NDVI) and normalized difference built-up index (NDBI). Pixels with $NDBI > 0.2$ and $NDVI < 0.2$ were classified as urban—representing zones with minimal vegetation and substantial impervious surfaces—while remaining pixels were labeled rural. These cutoff values are in accordance with established methodologies (Macarof & Statescu, 2017). The $NDVI < 0.2$ threshold is commonly adopted to distinguish non-vegetated or urban land cover; remote sensing research has affirmed its efficacy in mesic environments (Drori et al., 2020) and experimental mapping has shown that NDVI values below 0.2 correspond to areas lacking healthy vegetation (Hashim, 2019). Likewise, NDBI values above 0.2 are widely recognized as indicators of built-up surfaces in urban remote sensing applications (Malik, 2019). To evaluate the robustness of these thresholds, a sensitivity analysis was conducted by systematically varying the cutoff values across a plausible range ($NDVI = 0.15–0.30$; $NDBI = 0.15–0.30$). Urban–rural classifications and corresponding UHII values were recalculated for each threshold combination, and the results were compared with those from the baseline ($NDVI < 0.2$; $NDBI > 0.2$). The sensitivity analysis revealed that UHII values and the proportion of urban pixels remained relatively stable across the tested thresholds, while Jaccard similarity indices confirmed high spatial agreement between alternative threshold combinations and the baseline classification. Finally, classification accuracy was assessed using stratified sampling, with overall accuracy and Cohen’s Kappa coefficient employed as evaluation metrics. The assessment confirmed the reliability of the classification approach (Appendix 1).

3.5 Calculation of land surface temperature (LST)

LST was derived from the thermal infrared bands of the Landsat imagery. For Landsat 5, band 6 was utilized, while for Landsat 8, band 10 was employed. The radiance values were converted to temperature using the following Eq. 3:

$$LST = \frac{K_2}{\ln\left(\frac{K_1}{L_\lambda} + 1\right)} - 273.15 \quad (3)$$

where K_1 and K_2 are sensor-specific calibration constants, and L_λ represents the spectral radiance. This method aligns with the procedures outlined by Avdan & Jovanovska (Avand & Moradi, 2021).

3.6 Calculation of urban heat island intensity (UHII)

UHII was determined by computing the difference in mean LST between urban and rural areas within each Thana:

$$UHII = LST_{urban} - LST_{rural} \quad (4)$$

where:

LST_{urban} : Mean Land Surface Temperature in urban areas.

LST_{rural} : Mean Land Surface Temperature in rural areas.

This metric quantifies the temperature differential attributable to urbanization within each thana. The approach is consistent with methodologies employed in previous studies examining urban heat island effects (Kusuma et al., 2023).

3.7 Spatial-temporal analysis

This study employed Space Time Cube (STC) analysis in ArcGIS Pro to investigate trends in Urban Heat Island Intensity (UHII) across Bangladesh from 1990 to 2023, using a three-year interval. The “Defined Locations” option was used to aggregate both spatial and temporal data (Rahaman et. al., 2025), allowing for the examination of how temperature patterns in urban built-up areas evolved over time. Each section of the cube represents a specific geographic location and time period, enabling a structured analysis of long-term thermal trends.

To identify statistically significant trends in UHII, the Mann–Kendall test was applied. This non-parametric statistical test is widely used for detecting monotonic trends in time series data (Kendall & Gibbons, 1990; Mann, 1945). The test systematically compares pairs of time points, assigning a score of 1 if the UHII value increased, -1 if it decreased, and 0 if no change occurred. The overall trend is determined by summing these scores, where a total score close to zero suggests no significant trend. The statistical significance of the observed trend is evaluated using the z-score and p-value, where a low p-value indicates a statistically significant trend, and the sign of the z-score determines whether the trend is increasing or decreasing over time (Esri, 2021).

The test statistic S is calculated using the following formula 5:

$$S = \sum_{i=1}^{n-1} \sum_{j=i+1}^n a_{ij} \quad (5)$$

where,

$$a_{ij} = \text{sign}(x_j - x_i) = \text{sign}(R_j - R_i) = \begin{cases} 1 & x_i < x_j \\ 0 & x_i = x_j \\ -1 & x_i > x_j \end{cases} \quad (6)$$

The ranks of x_i and x_j time series are defined as R_i and R_j , respectively. Assuming the data are independent and identically distributed, the mean and variance of S statistics are,

$$E(s) = 0 \quad (7)$$

$$V_o(S) = \frac{n(n-1)(2n+5)}{18} \quad (8)$$

where n is the number of observations. This results in a reduction of the variance of S due to the existence of tied ranks (equal observations).

$$V_o^*(S) = \frac{n(n-1)(2n+5)}{18} - \sum_{j=1}^m \frac{t_j(t_j-1)(2t_j+5)}{18} \quad (9)$$

where m is the number of groups of tied ranks, each with t_j tied observation. The test iterates over the time series dataset and approximates the distribution of S , which becomes increasingly normal as the number of observations increases. This implies that the longer the study period, the higher the accuracy of trend detection and prediction.

3.8 Emerging hotspot analysis of urban heat island intensity (UHII)

The Space Time Cube (STC) framework was employed to analyze the spatiotemporal trends of Urban Heat Island Intensity (UHII) from 1990 to 2023, identifying areas where thermal hotspots emerged, persisted, or diminished over time (Esri, 2021; Getis & Ord, 1992; Ord & Getis, 1995). This approach employs Getis-Ord local statistics to analyze the reduction of variance from Eq. 9, using the following Eq. 10 to calculate the z-score (Ali et al., 2025):

$$G_i^* = \frac{\sum_{j=1}^n w_{ij}x_j - \bar{X} \sum_{j=1}^n w_{ij}}{S \sqrt{\frac{n \sum_{j=1}^n w_{ij}^2 - \left(\sum_{j=1}^n w_{ij}\right)^2}{n-1}}} \quad (10)$$

where x_j is the attribute value for feature j , w_{ij} is the spatial weight between feature i and j , and n is the total number of features. The calculation of $\bar{X}$ and S is followed by Eqs. 11 and 12 (Ali et al., 2025).

$$\bar{X} = \frac{\sum_{j=1}^n x_j}{n} \quad (11)$$

$$S = \sqrt{\frac{\sum_{j=1}^n x_j^2}{n} - (\bar{X})^2} \quad (12)$$

The Mann–Kendall test statistics were applied alongside STC to classify sixteen distinct spatial–temporal clusters, characterizing the persistence and intensity of UHII trends over time (Esri, 2021). These clusters were classified based on their statistical significance and temporal evolution. A New Hot Spot represents an area that has become statistically significant in the most recent time step. A Consecutive Hot Spot appears significant in at least two sequential time steps near the end of the study period but does not exceed 90% overall significance. Intensifying Hot Spots demonstrate significant clustering in more than 90% of the time steps, with an increasing trend in UHII. In contrast, a Persistent Hot Spot remains statistically significant in more than 90% of the time steps but does not exhibit an increasing or decreasing trend in intensity. A Diminishing Hot Spot indicates a statistically significant decline in clustering intensity over more than 90% of the time intervals. Sporadic Hot Spots are irregular, appearing significant only in the most recent time step, whereas Oscillating Hot Spots alternate between significant hot and cold spots but remain significant in less than 90% of intervals. A Historical Hot Spot was statistically significant in at least 90% of earlier intervals but has lost its significance in recent years.

Similarly, Cold Spots were categorized into New, Consecutive, Intensifying, Persistent, Diminishing, Sporadic, Oscillating, and Historical Cold Spots, reflecting spatial clusters of low UHII values with classification criteria analogous to hot spots. This detailed categorization provides valuable insights into the complex spatial–temporal variation of UHII across Bangladesh, offering critical implications for urban climate adaptation, sustainable planning, and environmental management.

3.9 Forest-based forecasting of urban heat island intensity (UHII)

To predict future Urban Heat Island Intensity (UHII) trends up to 2040, a forest-based forecasting model was implemented using the Random Forest (RF) algorithm. The Space Time Cube (STC) framework, which was employed to analyze spatiotemporal trends of UHII from 1990 to 2023, provided the input dataset for the RF forecasting. This approach was selected after comparison with linear regression and support vector regression, where RF demonstrated superior performance

in capturing non-linear relationships and minimizing overfitting. The historical UHII dataset (1990–2023) was used to train the model. Model development included hyperparameter tuning to optimize the number of trees, maximum tree depth, and minimum leaf size. The number of trees was fixed at 100, as empirical testing showed that model accuracy stabilized beyond approximately 80 trees, with no significant improvement in the coefficient of determination (R^2) or the root mean square error (RMSE), while computational cost continued to increase. Thus, 100 trees were considered an efficient and robust setting consistent with standard practice in Random Forest modeling.

The RF model constructs multiple decision trees and generates an ensemble prediction, mathematically represented in Eq. 13:

$$\hat{Y}_t = \frac{1}{M} \sum_{m=1}^M T_m(X_t) \quad (13)$$

where $\hat{Y}_t$ is the predicted UHII at time t , M denotes the total number of decision trees, T_m represents the m -th tree, and X_t corresponds to the input features for the given year. The dataset was divided into training (70%) and validation (30%) subsets, with model performance evaluated using the coefficient of determination (R^2) and the root mean square error (RMSE). The validation results indicated strong predictive performance, with high R^2 values and low RMSE, confirming the reliability of the UHII forecasts.

Although the Forest-Based Forecast tool in ArcGIS Pro does not provide traditional Random Forest variable importance scores or per-feature attribution, the interpretability of the model was evaluated through the temporal structure of the forecasting algorithm itself. Because the model relies exclusively on the UHII time series at each spatial location, interpretability is derived from understanding how historical patterns inform future predictions. The RF ensemble places greater weight on locations exhibiting strong temporal autocorrelation, persistent upward or downward UHII trajectories, and consistent hotspot or cold spot behavior in the Space Time Cube. Conversely, locations with highly variable or irregular UHII histories contribute less to stable predictions. This framework offers a transparent rationale for the resulting forecasts by demonstrating that projected UHII values emerge directly from the continuity, magnitude, and variability of historical UHII trends.

The dataset was divided into training (70%) and validation (30%) subsets, with model performance evaluated using the coefficient of determination (R^2) and the root mean square error (RMSE). The validation results indicated strong predictive performance, with a mean

forecast RMSE of 0.79 and a validation RMSE of 1.37, confirming the model's reliability in predicting UHII trends (Appendix 2). The consistently low error values and high overall accuracy across time steps demonstrate the robustness of the forecasting model and its suitability for assessing long-term UHII dynamics.

4 Results

Figure 3 illustrates a map derived from the Emerging Hotspot Analysis (EHSA) of Urban Heat Island Intensity (UHII), based on historical Landsat data from 1990 to 2023. The analysis reveals distinct spatial patterns across various thanas in Bangladesh, highlighting regions with persistent heat intensity trends. Both notable hotspots and cold spots are identified, reflecting the severity and consistency of UHII over time.

In Dhaka Division, the thanas of Dhaka District, such as Dhamrai, Savar, and Keraniganj, emerged as significant hotspots. Savar was identified as an intensifying hotspot, reflecting an increasing UHII trend, while Keraniganj demonstrated a persistent hotspot pattern, indicating a long-standing and stable heat concentration. Similarly, Narayanganj District, with thanas such as Araihaazar, Bandar, Rupganj, and Sonargaon, exhibited persistent hotspots, suggesting a sustained and growing urban heat island effect.

In Barisal Division, several thanas in Barguna District, including Amtali, Bamna, Betagi, and Patharghata, exhibited distinct cold spot patterns. Bamna and Betagi were categorized as persistent cold spots, with over 90% cold spot intensity, indicating a consistently cooler trend over time. Conversely, Amtali showed a sporadic cold spot pattern, signifying intermittent cooling trends. In Barisal District, thanas such as Agailjhara, Babuganj, and Banaripara were identified as persistent cold spots, while Gournadi and Muladi demonstrated sporadic cold spot patterns.

In Khulna Division, UHII patterns presented a combination of warming and cooling trends. Batiaghata exhibited an oscillating hotspot pattern, indicating fluctuating but persistent heat stress over time. Meanwhile, thanas such as Phultala and Dumuria showed diminishing cold spots, suggesting a gradual reduction in cooling effects, likely linked to increased urbanization and land-use changes. In Satkhira District, Assasuni, Debhata, and Kalaroa were identified as new hotspots, reflecting the recent intensification of UHII in these areas.

The Chattogram Division displayed significant variations in UHII trends. In Chattogram District, Anwara and Patiya emerged as new cold spots, suggesting recent urban or environmental changes contributing to localized cooling trends. In contrast, Hathazari demonstrated a consecutive hotspot pattern, indicating a steady

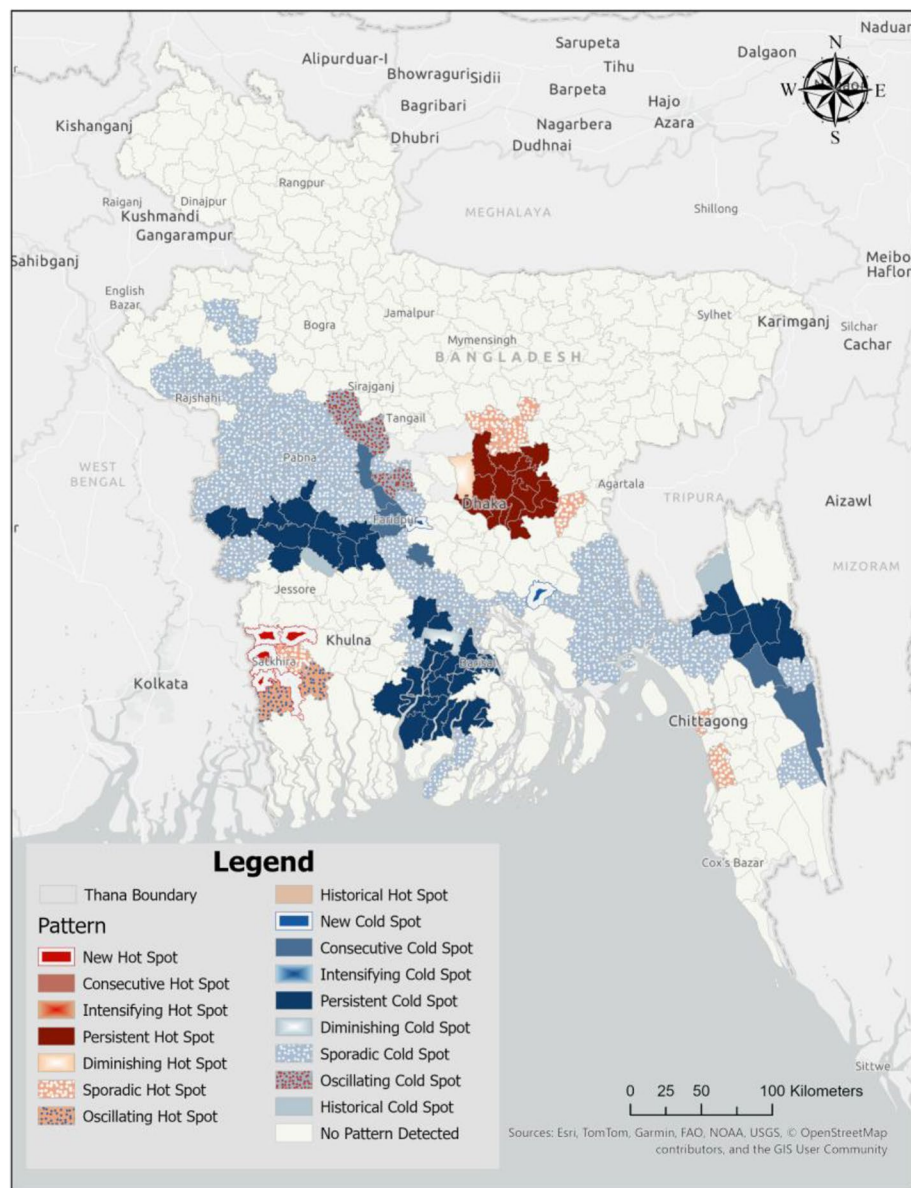


Fig. 3 Map illustrating the emerging hotspot analysis (EHSA) of urban heat island intensity (UHII) in Bangladesh (1990–2023)

increase in UHII over time. In Rangamati District, thanas such as Barkal, Langadu, and Nannerchar exhibited persistent cold spot patterns, reflecting long-term cooling trends.

In Rajshahi Division, Pabna District showed persistent cold spot patterns in thanas such as Bera and Sujaganar, highlighting regions that have maintained cooling trends over an extended period. Similarly, in Naogaon District, thanas such as Atrai and Mohadebpur exhibited sporadic cold spot patterns, indicating intermittent reductions in UHII intensity. In Sirajganj District, Shahjadpur and

Ullapara displayed oscillating cold spot patterns, suggesting localized fluctuations in temperature.

In Sylhet Division, Beanibazar and Golapganj thanas in Sylhet District emerged as new cold spots, reflecting recent land-use changes or climatic factors affecting UHII. Additionally, in Moulvibazar District, Sreemangal exhibited an oscillating cold spot pattern, further indicating variable but recurring cooling effects over time.

In Rangpur Division, the UHII patterns indicated localized reductions in temperature extremes. In Dinajpur District, Biral thana exhibited sporadic cold spots, suggesting intermittent cooling events rather than a

sustained trend. Meanwhile, in Thakurgaon District, Pirganj demonstrated a historical cold spot pattern, indicative of a prolonged cooling trend persisting over multiple decades.

The Mymensingh Division also displayed cooling trends in various thanas. In Mymensingh District, Trishal and Fulbaria thanas were identified as new cold spots, highlighting recent reductions in UHII intensity. Similarly, in Netrokona District, Madan and Mohanganj emerged as persistent cold spots, suggesting a stable and long-term cooling trend in these regions.

In general, the clustering of persistent hot spots in major urban centers such as Dhaka, Narayanganj, and

Khulna highlights the increasing intensity of the urban heat island effect, likely driven by rapid urbanization and infrastructural growth. Conversely, regions in Barisal, Rangpur, and Mymensingh Divisions exhibit persistent or emerging cold spot trends.

Figure 4 illustrates the spatial distribution of Urban Heat Island Intensity (UHII) in Bangladesh, derived from the Forest-Based Forecast Analysis projecting trends up to the year 2040. This projection is based on historical UHII data spanning from 1990 to 2023. The map reveals expected significant spatial patterns, predicting the emergence, persistence, and oscillation of both hotspots and cold spots across various Thanas.

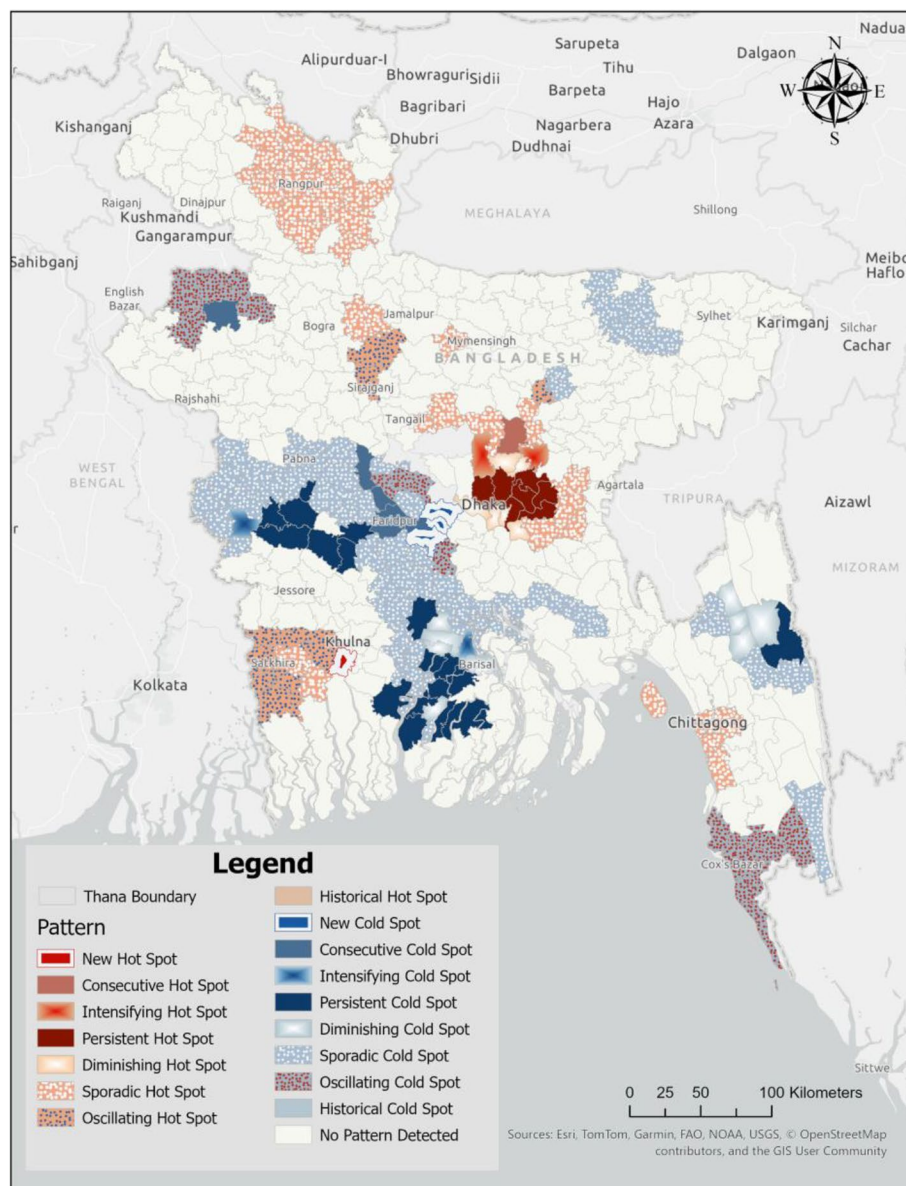


Fig. 4 Map depicting the projected spatial distribution of urban heat island intensity (UHII) in Bangladesh up to 2040, based on forest-based forecast analysis using 1990–2023 data

In Dhaka Division, the forecasted UHII trends indicate that Dhamrai, Savar, and Keraniganj in Dhaka District will continue to experience significant warming, with Savar projected as an intensifying hotspot, suggesting a worsening UHII effect. Similarly, Keraniganj is expected to sustain its status as a persistent hotspot, reinforcing long-term thermal stress in the area. The urban expansion in Narayanganj District also contributes to its forecasted persistent hotspot patterns in Arai-hazar, Bandar, Rupganj, and Sonargaon, emphasizing continued urban heat concentration.

In Barisal Division, several thanas in Barguna District, including Bamna, Betagi, and Patharghata, exhibit forecasted cooling trends. Betagi is projected as a persistent cold spot, reflecting stable and prolonged cooling, whereas Bamna shows a sporadic cold spot pattern, indicating intermittent reductions in UHII. Forecasts for Barisal District suggest contrasting trends, where Agailjhara and Banaripara are identified as diminishing cold spots, implying a gradual weakening of cooling effects. Meanwhile, Babuganj is projected as an intensifying cold spot, suggesting a strengthening cooling trend.

In Khulna Division, Batiaghata in Khulna District is forecasted to experience an oscillating hotspot pattern, suggesting variability in UHII trends over time. In contrast, thanas such as Phultala and Dumuria demonstrate diminishing cold spot patterns, indicating a gradual decrease in cooling intensity. In Satkhira District, the forecasted trends suggest a rise in UHII, with Assasuni, Debhata, and Kalaroa emerging as new hotspots, highlighting potential future urbanization impacts in these areas.

The Chattogram Division presents a mix of warming and cooling trends in the forecast. In Chattogram District, thanas such as Anwara and Patiya are projected to maintain new cold spot trends, suggesting localized cooling, whereas Hathazari is expected to retain its consecutive hotspot status, reinforcing sustained UHII intensification. Similarly, in Rangamati District, Barkal, Langadu, and Nannerchar are forecasted as persistent cold spots, indicating long-term cooling patterns likely influenced by topographical and ecological factors.

In Rajshahi Division, Pabna District exhibits projected persistent cold spot trends in thanas such as Bera and Sujanagar, signifying stable cooling effects. Similarly, in Naogaon District, Atrai and Mohadebpur are expected to retain their sporadic cold spot patterns, suggesting intermittent UHII reductions. In Sirajganj District, Shahjadpur and Ullapara are forecasted as oscillating cold spots, reflecting periodic fluctuations in temperature trends driven by land-use dynamics.

In Sylhet Division, Beanibazar and Golapganj in Sylhet District are projected as new cold spots, suggesting

emerging cooling effects. Similarly, Sreemangal in Moulvibazar District is expected to maintain its oscillating cold spot pattern, indicating dynamic shifts in UHII trends over time.

In Rangpur Division, forecasted cooling trends are evident in Dinajpur District, where Biral thana is projected to continue experiencing sporadic cold spots, suggesting periodic UHII reductions. Meanwhile, in Thakurgaon District, Pirganj is expected to sustain its historical cold spot status, indicative of a long-term cooling trend persisting into the future.

The Mymensingh Division is also projected to exhibit continued cooling trends. In Mymensingh District, Trishal and Fulbaria thanas are expected to emerge as new cold spots, reinforcing recent UHII reductions. In Netrokona District, Madan and Mohanganj are forecasted as persistent cold spots, suggesting stable and long-term cooling patterns.

Overall, the forest-based forecast emphasizes the continued intensification of urban heat effects in major urban centers such as Dhaka, Narayanganj, and Khulna, where persistent and intensifying hotspots are expected to remain dominant. Conversely, Barisal, Rangpur, and Mymensingh Divisions exhibit strong signals of cold spot persistence.

Figure 5 shows the rate of change in Urban Heat Island Intensity (UHII) across Bangladesh revealing significant spatial variations in hotspot dynamics. The rate of change is categorized into three primary trends: accelerating, decelerating, and stable, indicating the evolving UHII patterns at the Thana level.

In the Rangpur Division, Tetulia Thana in the Pan-chagarh District exhibits an accelerating trend in the Urban Heat Island Intensity (UHII), indicating an increasing heat intensity over time. In contrast, Atwari and Boda in the same district, along with Domar in the Nilphamari District, demonstrate a decelerating trend, suggesting a potential stabilization of urban heating effects.

In the Dhaka Division, several thanas show significant accelerating trends, emphasizing worsening UHII effects. Notably, areas within Dhaka District present strong evidence of UHII intensification, suggesting increased urbanization and land surface modifications. This aligns with the well-documented rapid urban expansion in the capital region.

A similar accelerating trend is evident in Khulna Division, particularly in Jessore District, where urban expansion and industrial activity might be contributing to sustained heat accumulation. Faridpur District, located in the Dhaka Division, also displays areas of UHII intensification, reinforcing concerns about the growing thermal footprint in central Bangladesh. Also, in Mymensingh

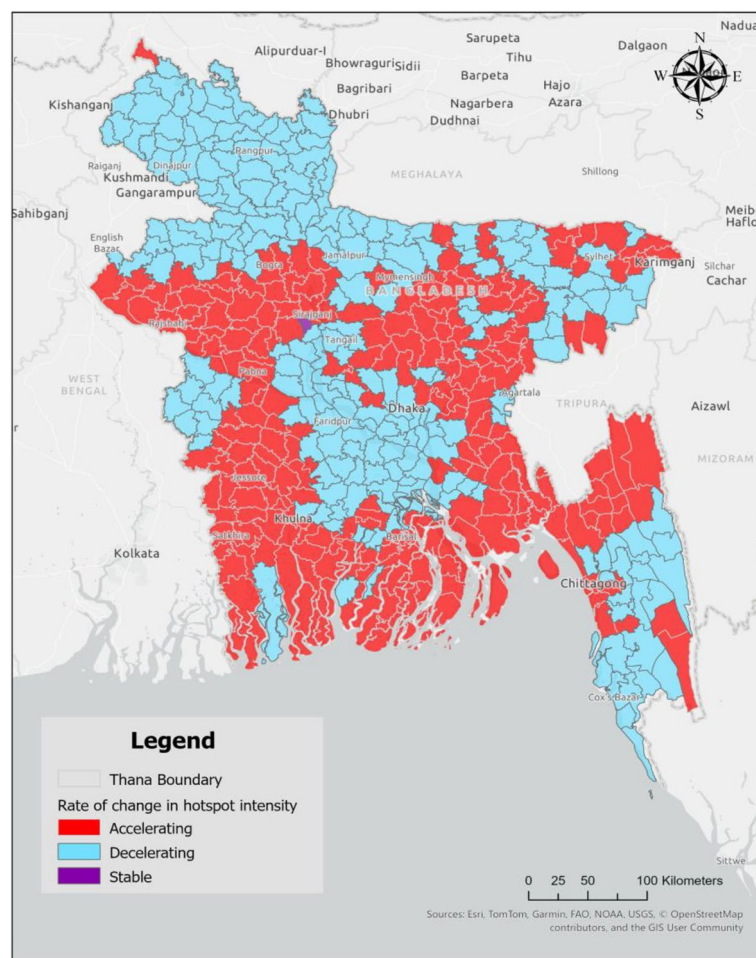


Fig. 5 Rate of change in urban heat island intensity (UHII) highlighting accelerating, decelerating, and stable trends at the thana level

Division, Mymensingh Sadar showed an accelerating trend, highlighting the region's growing urban heat concerns.

Conversely, in the Sylhet Division, Kurigram District demonstrates a decelerating trend, suggesting potential shifts in land-use or microclimatic influences mitigating excessive heat buildup. This contrasts with findings from the Chattogram Division, where Chattogram District shows an accelerating UHII pattern, likely driven by urban expansion and coastal development dynamics.

The presence of stable hotspots in select locations, such as Tangail District in Dhaka Division, suggests areas where UHII intensity has remained relatively unchanged over time. These stable patterns may indicate equilibrium between urban heating factors and potential mitigation measures.

Overall, the rate of change analysis underscores a complex interplay of urbanization, land use changes, and environmental factors influencing UHII dynamics across Bangladesh. The accelerating hotspots in densely

populated urban areas contrast with decelerating trends in less urbanized regions, emphasizing the need for targeted urban planning and climate adaptation strategies to mitigate the adverse impacts of UHII.

The transition matrix in Table 1 presents the spatial shifts in Urban Heat Island Intensity (UHII) patterns, comparing the Emerging Hotspot Analysis (EHSA) from 1990–2023 with the forecasted EHSA for 2040, highlighting notable changes in heat and cold spot dynamics across Bangladesh. This analysis highlights the extent of transitions in various patterns, measured in square kilometers (km^2), and provides critical insights into the evolving landscape of UHII.

A notable observation is the substantial transition from consecutive cold spots to areas with no pattern detected, accounting for approximately 945.73 km^2 . This shift suggests a potential weakening of previously stable cooling trends, indicating that regions once characterized by consistent cooling may experience more variable temperature patterns in the future. Additionally, about 674.10

Table 1 Transition matrix illustrating spatial shifts in UHII patterns between observed (1990–2023) and forecasted (2040) periods across Bangladesh

		Transition Matrix of UHII Patterns (Observed vs. Predicted)																
Original Pattern	Consecutive Cold Spot -	788.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	674.1	0.0	
	Consecutive Hot Spot -	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
	Diminishing Cold Spot -	0.0	0.0	248.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
	Diminishing Hot Spot -	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	347.9	0.0	0.0	0.0	0.0	0.0	
	Historical Cold Spot -	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	666.6	0.0	0.0	0.0	0.0	0.0	
	Historical Hot Spot -	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
	Intensifying Cold Spot -	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
	Intensifying Hot Spot -	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
	New Cold Spot -	136.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	233.0	
	New Hot Spot -	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1316.1	0.0	0.0	0.0	
	No Pattern Detected -	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	687.1	236.9	72408.7	5088.1	1572.8	0.0	0.0	4436.2	9043.3
	Oscillating Cold Spot -	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	957.9	354.6	0.0	0.0	0.0	0.0	0.0
	Oscillating Hot Spot -	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	441.3	0.0	0.0	0.0	385.2
	Persistent Cold Spot -	0.0	0.0	1889.4	0.0	0.0	0.0	430.5	0.0	0.0	0.0	497.1	0.0	0.0	5305.0	0.0	1567.5	0.0
	Persistent Hot Spot -	0.0	0.0	0.0	783.3	0.0	0.0	0.0	570.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1513.4	0.0
	Sporadic Cold Spot -	388.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	11485.3	0.0	0.0	0.0	0.0	9157.1	0.0
	Sporadic Hot Spot -	0.0	358.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1829.8
		Consecutive Cold Spot -	Consecutive Hot Spot -	Diminishing Cold Spot -	Diminishing Hot Spot -	Historical Cold Spot -	Historical Hot Spot -	Intensifying Cold Spot -	Intensifying Hot Spot -	New Cold Spot -	New Hot Spot -	No Pattern Detected -	Oscillating Cold Spot -	Oscillating Hot Spot -	Persistent Cold Spot -	Persistent Hot Spot -	Sporadic Cold Spot -	Sporadic Hot Spot -
		Transition To																

km² of consecutive cold spots have transitioned into sporadic cold spots, reflecting a reduction in the consistency of cooling effects.

For diminishing cold spots, around 248.86 km² transitioned to consecutive cold spots, indicating a re-emergence of stable cooling trends in certain regions. However, a significant portion of diminishing cold spots has also disappeared, with areas transitioning to no pattern detected, suggesting an overall decline in persistent cooling trends.

The dynamics of diminishing hot spots reveal a notable shift of 347.86 km² into areas with no detectable pattern, indicating that some previously warming regions may stabilize or exhibit less pronounced UHII effects. Similarly, historical cold spots covering 666.61 km² transitioned into no pattern detected, suggesting the loss of long-standing cooling effects in these areas. In contrast, the emergence of new cold spots is observed, with 136.66 km² transitioning from consecutive cold spots, indicating the development of localized cooling trends in new areas.

Furthermore, 232.99 km² of new cold spots transitioned from sporadic cold spots, highlighting the potential intensification of cooling effects in regions that previously experienced intermittent cooling.

The matrix also reveals limited transitions towards hot spot categories, with minimal areas transitioning into persistent or intensifying hot spots. This indicates that while certain regions are expected to experience increased UHII, the overall expansion of persistent heat-affected areas may be more localized. So, the transition matrix points out a complex interplay between cooling and warming trends, with significant areas losing their historical cooling patterns, while new cold spots emerge in different regions.

Figure 6 depicts urban heat island intensity (UHII) transitions highlighting significant spatial shifts from coldspot and no-pattern categories to hotspot categories across various divisions. The findings illustrate the increasing vulnerability of certain regions to intensified

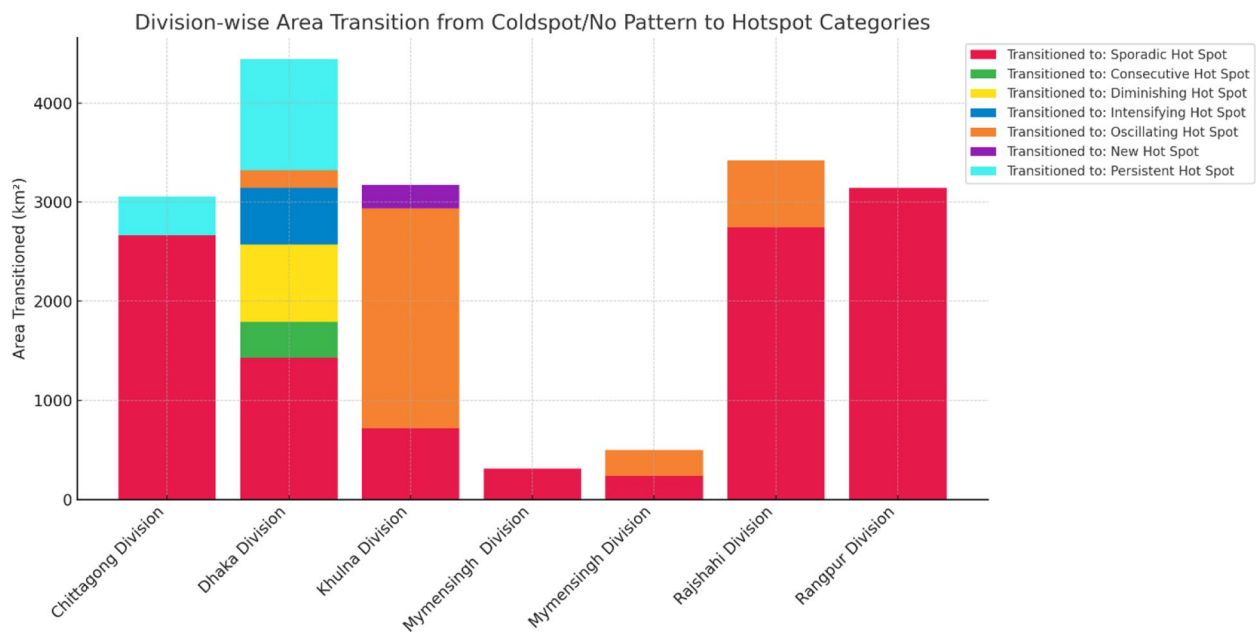


Fig. 6 Projected UHII transitions from coldspots/no-pattern to hotspots across divisions by 2040

heat stress, particularly in urbanizing and industrially expanding areas.

The most notable transitions are observed in Dhaka Division, where a significant area previously categorized as "No Pattern Detected" and coldspots has transitioned into Intensifying Hot Spots (570.75 km²), Diminishing Hot Spots (783.33 km²), and Persistent Hot Spots (1121.27 km²). This transformation indicates a strong warming trend. The emergence of Sporadic Hot Spots (1430.45 km²) further suggests the localized intensification of heat stress in dynamically changing urban and peri-urban environments.

A similar pattern is evident in Chattogram Division, where Sporadic Hot Spots (2664.66 km²) have emerged over a considerable area, suggesting an increasing frequency of temporary yet significant heat events. The division also exhibits a transition toward Persistent Hot Spots (392.17 km²), indicating sustained warming trends. In Khulna Division, a significant portion of the area has transitioned into Oscillating Hot Spots (2216.63 km²) and New Hot Spots (236.92 km²). These transitions suggest fluctuations in UHII intensity over time.

The Mymensingh Division shows relatively smaller transitions compared to Dhaka and Chattogram, with Sporadic Hot Spots emerging in approximately 311.43 km². This moderate shift suggests that while certain areas are experiencing heat intensification, the overall trend remains less pronounced than in highly urbanized divisions. So, the observed spatial shifts emphasize the dynamic nature of UHII patterns, with significant areas

that were previously categorized as coldspots or no-pattern zones now exhibiting hotspot characteristics.

5 Discussion

The findings of this study reveal a complex and evolving pattern of Urban Heat Island Intensity (UHII) across Bangladesh, with distinct spatial and temporal trends observed at the Thana level. The identification of persistent and intensifying hotspots, particularly in major urban centers like Dhaka, Narayanganj, and Khulna, highlights the profound impact of rapid urbanization, land-use changes, and population growth on local thermal environments. Conversely, the presence of emerging and persistent cold spots in divisions like Barisal, Rangpur, and Mymensingh suggests that specific climatic and land-cover dynamics may be mitigating the effects of urban heating in these areas.

5.1 Drivers of UHII intensification in Bangladesh

The significant increase in UHII in urban regions, particularly in Dhaka, Narayanganj, and Khulna, can be attributed to expanding built-up areas, loss of vegetation, and increased anthropogenic heat emissions. The growth of impervious surfaces, such as concrete and asphalt, has led to an increase in heat absorption and a reduction in evapotranspiration, intensifying daytime temperatures and limiting nighttime cooling (Santamouris, 2020). These findings align with previous studies that highlight the strong correlation between urban expansion and rising land surface temperatures (LST) (Pal & Ziaul, 2017).

Furthermore, industrialization and transportation activities in these regions contribute significantly to localized warming. Emissions from vehicles, manufacturing units, and commercial activities release waste heat, exacerbating the urban thermal load (Wang et al., 2016). The increasing UHII effect in these areas aligns with global trends where dense urban landscapes experience higher temperature anomalies due to the cumulative impact of anthropogenic heat sources and reduced natural cooling mechanisms (Santamouris, 2015).

Another key factor driving UHII intensification is the decline of vegetative cover, as evident from NDVI trends in the study area. Vegetation plays a critical role in mitigating urban heat through shading and evapotranspiration, which help cool urban surfaces (Rahman et al., 2021). The loss of green spaces in major cities like Dhaka and Chattogram aligns with prior research indicating that a reduction in urban green infrastructure leads to a substantial rise in LST and an overall intensification of UHII (Abdullah et al., 2022). The continued expansion of high-density residential and commercial zones, coupled with inadequate urban greening initiatives, has likely contributed to the observed increase in UHII in these metropolitan regions.

5.2 Cooling trends and emerging cold spots

While urban centers are experiencing intensifying UHII, notable cold spots have emerged in regions such as Barisal, Rangpur, and Mymensingh, reflecting areas where cooling trends persist. These findings suggest that localized climatic conditions, land cover composition, and geographical features may be playing a role in moderating heat accumulation in these areas.

One possible explanation for these persistent cold spots is the presence of large water bodies and dense vegetation cover, which enhance cooling through evapotranspiration and moisture retention (Gohain et al., 2021). Coastal areas in Barisal, for example, benefit from the Bay of Bengal's maritime influence, which contributes to localized cooling and reduces temperature extremes (Ahmed et al., 2013). Similarly, forested regions in Mymensingh and Rangpur serve as natural heat sinks, mitigating urban temperature increases (Dewan et al., 2021).

Additionally, differences in urban morphology and development intensity may also explain the spatial variation in UHII trends. Less industrialized and lower-density urban regions exhibit milder heat island effects due to the presence of mixed land use patterns and fewer impervious surfaces, allowing for greater airflow and natural cooling (Leal Filho et al., 2021). This supports findings from global studies that emphasize the role of urban planning and land-use diversity in moderating urban heat stress (Santamouris, 2020).

5.3 Projected UHII trends and future risks

The forecasted UHII trends up to 2040 indicate a worsening of urban heat stress in densely populated and industrially expanding regions, particularly in Dhaka, Narayanganj, and Chattogram. These projections align with global climate models that predict an increase in urban extreme heat events due to climate change and uncontrolled urban expansion (Zhao et al., 2021).

One of the most concerning aspects of this projection is the emergence of new and intensifying hotspots, highlighting the potential for increasing thermal discomfort, rising cooling energy demands, and public health risks (Mitchell et al., 2016; van Oldenborgh et al., 2018). Vulnerable populations, particularly outdoor workers, the elderly, and those in informal settlements, will likely bear the brunt of worsening UHII impacts (Chakraborty et al., 2019; Heaviside et al., 2017).

Moreover, the forecasted UHII expansion into peri-urban areas suggests that unplanned urban sprawl could further exacerbate heat-related challenges in rapidly developing zones. This underscores the need for climate-sensitive urban policies, including the integration of urban forestry, reflective surfaces, and green infrastructure, to mitigate the future impact of UHII (Santamouris, 2020).

6 Limitations

While this study presents a comprehensive assessment of Urban Heat Island Intensity (UHII) trends across Bangladesh using advanced remote sensing and machine learning techniques, several limitations should be acknowledged.

First, data resolution constraints pose challenges in capturing micro-level variations in UHII. The study relies on Landsat thermal imagery with a 30-m spatial resolution, which is suitable for broad-scale analysis, but may not fully capture localized heat variations within densely populated urban settings. Higher-resolution thermal data from sources such as ECOSTRESS or commercial satellite platforms could enhance spatial precision in future studies.

Second, the study treats land surface temperature (LST) as the primary indicator of UHI; however, LST captures only one dimension of the UHI phenomenon. While LST is valuable for detecting heat retention and emission patterns, it does not adequately reflect ambient air temperatures, which play a critical role in shaping the overall UHI effect. These atmospheric factors are central to canopy-layer UHI processes and human thermal comfort. The absence of air temperature and humidity data, particularly from ground-based meteorological or microclimatic observations, limits the ability to fully validate satellite-derived UHII estimates and to capture

the human-perceived intensity of urban heat. Moreover, LST-derived UHII estimates inherently carry uncertainty due to emissivity assumptions, atmospheric correction procedures, and sensor-specific radiometric noise. A more explicit quantification of these uncertainties, particularly how they propagate into UHII calculations, would strengthen the reliability of future assessments.

Third, land cover classification uncertainties may influence UHII estimations. Although the study employs the Normalized Difference Vegetation Index (NDVI) and Normalized Difference Built-up Index (NDBI) for urban–rural delineation, potential misclassifications due to seasonal vegetation changes and spectral mixing can introduce errors. Additionally, while the study conducts an accuracy assessment, it does not incorporate land use–LST overlay analysis, which could have provided a more explicit evaluation of how land cover transitions contribute to UHII dynamics. Acknowledging this limitation is essential, as land-use overlays can reveal spatial inconsistencies between thermal hotspots and physical landscape features that single-threshold indices may not fully capture. Future studies should integrate multi-temporal land use layers with LST fields to better contextualize UHII patterns and reduce classification-related uncertainty.

Fourth, socioeconomic and demographic factors were not explicitly incorporated into the analysis. Urban heat exposure is not solely dictated by land cover changes but is also influenced by population density, building materials, urban design, and socioeconomic disparities. The integration of census data, energy consumption patterns, and urban morphology indicators could provide a more holistic understanding of UHI impacts on human well-being.

Lastly, machine learning model uncertainties present limitations in UHII forecasting. While Random Forest (RF) was chosen for its robustness and interpretability, alternative predictive models such as Long Short-Term Memory (LSTM) networks or hybrid deep learning approaches could further refine UHII projections by incorporating non-linear temporal dependencies.

7 Concluding remarks

The findings of this study highlight the intensifying Urban Heat Island Intensity (UHII) in Bangladesh, particularly in rapidly urbanizing regions, where the effects of land use changes and anthropogenic heat emissions are becoming more pronounced. The spatial and temporal analysis highlights how urban expansion continues to exacerbate localized warming, with major metropolitan centers experiencing persistent and intensifying heat stress. This trend aligns with global

patterns but presents unique regional challenges, particularly in the context of Bangladesh's high population density, limited green spaces, and infrastructural constraints. The results point out the burning need for integrating urban climate considerations into planning frameworks to mitigate the adverse effects of UHI on public health and ensure a sustainable environment.

Given the projected increase in UHII, this study should be regarded as an exploratory approach that identifies emerging trends and serves as a pilot for more detailed future investigations. Future research will focus on developing targeted mitigation strategies for identified hotspot areas and linking them with underlying urban typologies, land use characteristics, and environmental drivers. In particular, future research plans to integrate land use classification data to determine which specific land use types contribute most to UHII intensification at the thana level, thereby enhancing the spatial precision and policy relevance of the findings. Specific adaptation measures, such as cool roofs in Dhaka, permeable pavements in Chattogram, and green corridors in Khulna, should be evaluated alongside cost–benefit analyses to guide sustainable urban planning decisions. A more comprehensive understanding of land–atmosphere interactions is also essential, particularly in quantifying the effects of industrial heat emissions, energy consumption, urban morphology (e.g., sky view factor), and localized wind circulation. Furthermore, integrating higher-resolution remote sensing datasets, census-derived socio-economic indicators (e.g., population density, income levels, and health metrics), and real-time climate monitoring networks will significantly improve both the accuracy and equity of future urban heat risk assessments.

On the modeling side, predictive frameworks should be expanded beyond Random Forest to include explainable machine learning approaches, such as SHAP values for feature attribution, and alternative models like XGBoost or LSTM for robustness and temporal forecasting. Such approaches will enhance interpretability and transparency, ensuring that the drivers of projected UHII patterns are better understood. Future studies should also address urban heat disparities by examining how low-income communities, informal settlements, and outdoor workers are disproportionately exposed to UHII compared to more affluent areas. This requires a policy-driven approach focused on equitable distribution of cooling infrastructure, climate-conscious zoning, and community-driven urban renewal projects.

As Bangladesh continues its urban transformation, balancing development needs with climate resilience will be imperative. The study findings provide a foundation for evidence-based urban climate policies, yet there remains

a critical need for localized and community-driven adaptation strategies. Strengthening institutional capacity, increasing public awareness, and promoting a culture of sustainable urban design will be key in ensuring that cities remain habitable, energy-efficient, and climate-resilient in the coming decades (United Nations Office for Disaster Risk Reduction, (n.d.)). Future research must build on these insights to bridge knowledge gaps and support integrated urban climate management, ensuring that scientific advancements translate into practical solutions for mitigating the growing threat of UHI in Bangladesh.

Appendix

Year-wise accuracy assessment of UHII classification (1990–2023)

Year	Kappa	Overall accuracy
1990	0.8426	0.9596
1993	0.9643	0.9877
1996	0.8205	0.9510
1999	0.8840	0.9680
2002	0.7916	0.9440
2005	0.9071	0.9735
2008	0.8890	0.9683
2011	0.9710	0.9914
2014	0.8842	0.9707
2017	0.9000	0.9539
2020	0.8543	0.9537
2023	0.9603	0.9895

Summary of forest-based forecasting results derived from the UHII Space–Time Cube analysis (1990–2023)

Parameter	Value
Study Period	1990–2023
Time Step Interval	3 years
Forecast Method	Forest-Based
Locations Analyzed	459
Space–Time Bins	5508
Forecast RMSE (Mean $\pm$ SD)	0.79 $\pm$ 0.43
Validation RMSE (Mean $\pm$ SD)	1.37 $\pm$ 0.98
Forecast Period	2023–2041

Authors' contributions

Al Artat Bin Ali: Conceptualization, Methodology, Software, Formal Analysis, Writing Original Draft, Revision; Chandana Mitra: Conceptualization, Supervising; Faiyad H Rishal: Reviewing and editing, Revision; Rifat Bin Hossain: Writing Original Draft, Revision.

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Data availability

The data presented in this study are available on request from the corresponding author.

Declarations

Competing interests

On behalf of all authors, the corresponding author states that there is no conflict of interest.

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